Knowledge as Normalization: Bounded Context Separates Shared Propositions from Action

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Abstract

When multiple action policies share propositions and active context is bounded, storing each policy's propositions independently produces $O(n \cdot k)$ context load, multiplicative in the number of policies n and propositions per policy k. We prove that bounded context forces the extraction of shared propositions into a separate structure K, reducing context load to $O(n + k')$ where $k' = |K|$. This is knowledge as normalization: K precipitates from the action space A the same way normal forms precipitate from redundant relations in Codd's relational model. The normalization is not a design choice but a scaling constraint imposed by bounded C_n. From the normalization theorem we derive the retention criterion (retain when benefit is uncertain; prune only when benefit is certainly zero) and K/A inseparability (the content of K is constitutively determined by A). These results engage with recent proposals to give observation and action separate training objectives [6]. Such separation violates a structural constraint imposed by bounded context.

1 Introduction

When a tumor severs a coupling channel, the cell loses both the information that channel carried and the actions that information supported [21]. A cell that can no longer sense tissue-level growth signals can no longer execute tissue-coordinated quiescence. Knowledge and action are inseparable: losing one entails losing the other. The present paper proves that this inseparability is a structural consequence of bounded context.

McCarthy [13] establishes that any entity accumulating propositions under bounded active context necessarily maintains a directed graph with typed edges over its propositions (where "necessary" means selected for under competition—see [13], Remark 1). That result answers a structural question: *what form* must knowledge take? McCarthy [14] establishes that control architectures converge to escalation under bounded context. The present paper asks the next question: *why does a separated knowledge structure emerge at all?*

The answer is a scaling constraint. Consider an entity with n action policies, each requiring k propositions on average, with overlap fraction β (the proportion of propositions shared across two or more policies). Without a separated knowledge structure, each policy carries its own copy of every required proposition. Total context load is $O(n \cdot k)$. Under bounded active context C_n, this architecture fails for any non-trivial repertoire: two overlapping policies loaded simultaneously require $2k - |\text{overlap}|$ context slots, and without separation the overlap is not recognized: both full copies are loaded.

With a separated structure K, shared propositions are stored once and referenced by multiple policies. Total context load drops to $O(n + k')$ where $k' = |K|$ absorbs the shared propositions. The per-action marginal cost approaches $O(1)$ as overlap increases.

This is Codd's insight [1] applied to bounded cognition. Codd showed that redundant copies across relations produce update anomalies and wasted storage; normal forms extract shared dependencies into referenced tables. The same pressure operates here: bounded context is the capacity constraint, and K is the normal form.

The paper proves a single theorem, the Normalization Theorem, and derives corollaries establishing a retention criterion for propositions, the inseparability of K and A, and the asymptotic compactness of K toward action-relevant propositions. These results engage prior work on ecological perception [11], bounded rationality [18], the information bottleneck [19, 20], and active inference [8, 9].

2 Formal Framework

Definitions 1 and 2 recapitulate [13]; Definitions 3–7 are new.

Definition 1 (World State Space)**.** Let W be a measurable space of possible world states. At time t, the true state $w_t \in W$ is not directly observable by any entity.

Definition 2 (Propositions and Posterior)**.** A *proposition* p about W is a likelihood function $\ell_p : W \to [0, 1]$. A set of propositions $\mathcal{P}$ induces a posterior via the joint likelihood $L_{\mathcal{P}}(w)$:

$$P(w \mid \mathcal{P}) \;\propto\; P_0(w) \cdot L_{\mathcal{P}}(w)$$

where P_0 is a prior and $L_{\mathcal{P}}$ encodes the joint evidential contribution of all propositions in $\mathcal{P}$, with arbitrary dependencies between them (Paper 1, [13]). When propositions are conditionally independent given w, $L_{\mathcal{P}}(w) = \prod_{p \in \mathcal{P}} \ell_p(w)$; the normalization theorem does not require this special case. The uncertainty about w is the surprisal $U(w, \mathcal{P}) = -\log P(w \mid \mathcal{P})$ [17, 3]. We use K to denote the separated set of shared propositions that Theorem 1 derives from the overlap structure of action policies. At the definitional stage, K is not assumed to exist; the theorem establishes its necessity.

Definition 3 (Action Space)**.** Let Ω be a set of executable actions and $W_{\text{obs}} \subseteq W$ the observable component of the world state. The *action space* is

$$A \;\subseteq\; (\mathcal{P} \times W_{\text{obs}} \to \Delta\Omega)$$

where $\Delta\Omega$ is the space of distributions over Ω and $\mathcal{P}$ is the entity's proposition pool (Definition 2). Each $a \in A$ selects an action distribution given propositions and current observations. For each a, the *grounding set* $\mathcal{P}^{[a]} \subseteq \mathcal{P}$ is the minimal subset of $\mathcal{P}$ such that a's output is fully determined by $\mathcal{P}^{[a]}$ and W_{obs}. Theorem 1 shows that the shared propositions across action policies separate into a structure K; after that separation, we write $K^{[a]}$ for the grounding set within K.

Action policies span a spectrum of propositional dependence. When $\mathcal{P}^{[a]} = \emptyset$, a depends only on W_{obs}, a reflex or stimulus-response mapping that requires no propositional knowledge. When $|\mathcal{P}^{[a]}| > 0$, a is *knowledge grounded*: its output depends on stored propositions.

Definition 4 (Entity)**.** An entity is a tuple $E = (K, A, \sigma, \Omega)$ where K is the knowledge graph (Definition 2), A is the informed action space (Definition 3), $\sigma : W \to W_{\text{obs}}$ is the sensing function, and Ω is the set of executable actions.

Definition 5 (Bounded Active Context)**.** C_n denotes the active context capacity: the maximum number of propositions simultaneously available for inference. Let $\mathrm{ctx}(t) \subseteq \mathcal{P}$ be the propositions in active context at time t, with $|\mathrm{ctx}(t)| \leq C_n$. The relevance density for action policy a is:

$$\rho(a) = \frac{|\mathcal{P}^{[a]} \cap \mathrm{ctx}(t)|}{|\mathrm{ctx}(t)|}$$

Definition 6 (Survival Threshold)**.** For each domain d, let U^d_{lethal} be the uncertainty threshold above which the entity does not survive: if $U(w, K) > U^d_{\text{lethal}}$ for the relevant world states in domain d, the entity's continued existence is not assured.

Definition 7 (Information Gain)**.** When the entity executes action a and receives feedback encoded as proposition p_a, the information gain is:

$$G(a, K) = I(p_a; W \mid K) \;\geq\; 0$$

where $I(\cdot\,;\cdot \mid \cdot)$ is conditional mutual information [3].

3 The Normalization Theorem

The theorem establishes that bounded active context forces the separation of shared propositions from action policies.

Definition 8 (Context Load)**.** For an entity with action space A and knowledge representation $\mathcal{R}$, the *context load* $L(\mathcal{R}, A)$ is the number of proposition slots required to simultaneously evaluate any pair of action policies.

Theorem 1 (Normalization of Shared Propositions)**.** *Let $E = (K, A, \sigma, \Omega)$ be an entity with bounded active context C_n (Definition 5), $n = |A|$ knowledge-grounded action policies, average proposition count k per policy, and overlap fraction $\beta > 0$, defined over the raw proposition pool: $\beta = |\{p \in \mathcal{P} : |\{a \in A : p \in \mathcal{P}^{[a]}\}| \geq 2\}| \,/\, |\mathcal{P}|$, the proportion of propositions referenced by two or more policies. This is well-defined before K exists, since it depends only on $\mathcal{P}$ and the grounding sets $\mathcal{P}^{[a]}$. Then:*

(N1) ***Unnormalized failure.*** *Under an implicit-K architecture where each policy carries its own propositions, the context load for evaluating two overlapping policies is $L_{impl} = 2k - |overlap|$, and total proposition storage is $n \cdot k$. For any pair (a_1, a_2) such that $L_{impl} > C_n$, the architecture cannot simultaneously evaluate both policies. More generally, as n grows, the fraction of policy pairs that exceed C_n increases, progressively restricting the entity's ability to compare and select actions.*

(N2) ***Normalized architecture.*** *Extracting shared propositions into a separated structure K with each policy referencing shared propositions via graph edges produces context load $L_{norm} = |K^{[a_1]} \cup K^{[a_2]}|$ for any pair (a_1, a_2), with total storage $|K| + n \cdot (1 - \beta) \cdot k$. Each shared proposition is loaded once regardless of how many policies reference it.*

(N3) ***Scaling.*** *$L_{norm} \leq L_{impl}$ with strict inequality whenever $\beta > 0$. In the limit $\beta \to 1$, total storage approaches $|K| + n$ and per-action marginal context cost approaches $O(1)$.*

(N4) ***Anomaly elimination.*** *Under implicit-K, updating a shared proposition p requires modifying n copies, producing the update anomaly: one copy is revised while others retain the stale value, yielding inconsistent action policies. Under normalized K, updating p is $O(1)$ and all referencing policies see the revision immediately.*

Therefore, under bounded C_n with $\beta > 0$, the separated K architecture is necessary for scaling the action repertoire beyond C_n/k policies.

Proof. The proof proceeds by counting context slots under each architecture.

Part (N1): Unnormalized context load. Under the implicit-K architecture, each action policy a_i stores its own copy of every proposition it requires: the set K_i^{local} with $|K_i^{\text{local}}| = k$. When two policies a_1, a_2 must be evaluated simultaneously (e.g., to compare expected outcomes), the entity must load K_1^{local} and K_2^{local} into active context. Since the copies are independent, the overlap between K_1^{local} and K_2^{local} is not recognized by the retrieval system: both full copies are loaded, requiring $2k$ context slots minus any structural deduplication the system performs. In the worst case (no deduplication), $L_{\text{impl}} = 2k$. With perfect structural deduplication, $L_{\text{impl}} = 2k - |\text{overlap}|$. In either case, total storage across all policies is $n \cdot k$, and when $n \cdot k > C_n$, the entity cannot represent its full repertoire.

Part (N2): Normalized context load. Under the normalized architecture, shared propositions are stored once in K and each policy a_i is represented as a function of $K^{[a_i]} \subseteq K$ and W_{obs}. When two policies a_1, a_2 are evaluated simultaneously, the entity loads $K^{[a_1]} \cup K^{[a_2]}$ into context. Shared propositions in $K^{[a_1]} \cap K^{[a_2]}$ are loaded once. Context load is $|K^{[a_1]} \cup K^{[a_2]}| = |K^{[a_1]}| + |K^{[a_2]}| - |K^{[a_1]} \cap K^{[a_2]}|$. Total storage is $|K| + n \cdot (1-\beta) \cdot k$, where the second term accounts for policy-specific (unshared) propositions.

Part (N3): Scaling comparison. $L_{\text{norm}} = |K^{[a_1]} \cup K^{[a_2]}| \leq |K^{[a_1]}| + |K^{[a_2]}|$, with strict inequality when $K^{[a_1]} \cap K^{[a_2]} \neq \emptyset$ (i.e., $\beta > 0$). In the limit $\beta \to 1$, nearly all propositions are shared: $|K^{[a_i]}|$ is dominated by K, and the per-policy marginal storage (the unshared component) approaches $O(1)$, giving total storage $|K| + n \cdot O(1) = O(|K| + n)$.

Part (N4): Update anomaly. Under implicit-K, a shared proposition p appears in K_i^{local} for every policy a_i that uses it. Updating p (e.g., upon receiving new evidence) requires identifying and modifying all n copies. If any copy is missed, a_i operates on stale information while a_j operates on the revision. This is exactly Codd's update anomaly [1]: redundant storage of the same fact causes inconsistency upon partial update. Under normalized K, p appears once; updating it is $O(1)$ and all policies referencing p via graph edges see the update immediately.

Necessity. The separation is not a design preference but a scaling constraint. Any entity with $n \cdot k > C_n$ and $\beta > 0$ that does not extract shared propositions will either (a) exceed its context bound, or (b) restrict its action repertoire to $\lfloor C_n/k \rfloor$ policies. Since survival pressure requires maintaining action policies across multiple domains (Definition 6), and domain coverage grows with n, the context bound forces normalization for any entity whose action repertoire is not trivially small. □

4 Corollaries

The normalization theorem establishes *that* shared propositions separate. The corollaries establish *what determines their content.*

4.1 Retention criterion

Once K exists as a separated structure, the question becomes: which propositions does K retain? The answer follows from the cost asymmetry inherent in bounded context under survival pressure.

Lemma 2 (Proposition Absence Cost)**.** *Let $p \in K$ be a proposition grounding some action policy $a \in A$ in domain d. Then:*

(i) Removing p from K weakly increases expected uncertainty: $H(W \mid K \setminus \{p\}) \geq H(W \mid K)$.

(ii) If p is the sole grounding for a critical action in domain d, then removal may cause $U(w, K \setminus \{p\}) > U^d_{lethal}$ for survival-relevant world states.

(iii) The cost of absent p when needed, C_{absent}, is bounded below by entity non-survival and therefore dominates all finite operational costs.

Proof. **Part (i).** Within the entity's model, the posterior is $P(w \mid K) \propto P_0(w) \cdot L_K(w)$, where L_K incorporates the constraints from all propositions in K. Removing proposition p removes a constraint: $L_{K\setminus\{p\}}$ is less restrictive than L_K, so the posterior is weakly broader and $H(W \mid K \setminus \{p\}) \geq H(W \mid K)$. This is model-internal monotonicity: it holds because the entity treats its own propositions as valid constraints.

Model vs. reality. Model-internal monotonicity does not guarantee that the entity's model tracks reality. A proposition may be wrong: the likelihood function may be mis-specified, the evidence may be noise, or the world may have shifted since the evidence was collected. In such cases, the entity's model entropy decreases (it becomes more confident) while its true uncertainty about W may increase. The retention criterion (Corollary 3) operates on model entropy because that is all the entity has access to; it cannot step outside its own model to evaluate its propositions against ground truth. Whether model entropy tracks true entropy over time is a question about learning quality, not about retention logic: an entity whose learning mechanism has positive expected accuracy (the propositions it acquires are more often right than wrong) will see its model converge toward reality; an entity with systematically misleading propositions will not. Paper 4 [15] formalizes this as the requirement $\mathbb{E}_t[\eta_M] > 0$. The retention criterion is correct given the entity's model; the model's fidelity to reality is a separate property.

Part (ii). If p is the sole grounding for the unique policy maintaining $U(w, K) \leq U^d_{\text{lethal}}$, removing p renders that policy ungrounded. No other policy suffices by assumption.

Part (iii). When $U > U^d_{\text{lethal}}$, the entity does not survive in domain d. Non-survival terminates all future utility, so $C_{\text{absent}} \gg c_{\text{retain}}$ for any finite retention cost. □

Corollary 3 (Retention Criterion)**.** *Let E be an entity with bounded C_n and normalized K (Theorem 1). The retention criterion for propositions in K is asymmetric:*

(R) ***Retention****: p is retained when there exists $a \in A$ applicable to p, or when the survival benefit of p is uncertain.*

(P) ***Pruning****: p is pruned only when the survival benefit of p is certainly zero across all reachable future trajectories.*

(D) ***Displacement*** *(curation regime, $|K| = K_{\max}$): p is displaced by candidate q when q has strictly higher net marginal value: expected survival benefit minus maintenance cost.*

Proof. Define the option value of retaining p:

$$\mathrm{OV}(p) = P\big(\exists\, \tau^* : B(p, \tau^*) > 0\big) \;\cdot\; \mathbb{E}\big[B(p, \tau^*) \mid B(p, \tau^*) > 0\big]$$

where $B(p, \tau^*)$ is the benefit of having p at future time τ^* along some reachable trajectory. This is a real option in the sense of Dixit and Pindyck [5]: retaining p preserves the option to act on p-grounded policies.

The net value of retaining p is $\mathrm{NV}_{\mathrm{retain}} = \mathrm{OV}(p) - c_{\mathrm{retain}}(p, K)$, where c_{retain} includes retrieval interference, edge maintenance, and context consumption. The net value of pruning is $\mathrm{NV}_{\mathrm{prune}} = -c_{\mathrm{prune}} - P(\exists\, \tau^*) \cdot C_{\mathrm{absent}}$.

By Lemma 2(iii), $C_{\mathrm{absent}} \gg c_{\mathrm{retain}}$ whenever p grounds any survival-relevant action. For propositions with uncertain relevance, the expected cost of wrongly pruning (C_{absent}, unbounded) dominates the expected cost of wrongly retaining (c_{retain}, bounded). Therefore:

$$\mathrm{NV}_{\mathrm{retain}} > \mathrm{NV}_{\mathrm{prune}} \quad \text{whenever} \quad P\big(\exists\, \tau^* : B(p, \tau^*) > 0\big) > 0$$

Retention dominates whenever there is any positive probability of future benefit. Pruning is correct only when $P(\exists\, \tau^*) = 0$, certain zero benefit, which is unverifiable for most propositions. In the curation regime ($|K| = K_{\max}$), every addition requires displacement by marginal value comparison.

The richer cost structure sharpens the result. For critical propositions, C_{absent} dominates c_{retain} regardless of maintenance cost. For non-critical propositions, as $|K|$ grows, interference and edge costs increase, raising the retention threshold. When $c_{\mathrm{retain}}(p, K) > \mathrm{OV}(p)$, displacement is correct: the entity replaces p with a candidate of higher net value. "Aggressive forgetting" in bounded systems is displacement, not pruning.

Note: the entity cannot classify propositions into "survival-relevant" and "not" with certainty. It can form imperfect ordinal rankings of expected marginal value. The retention criterion (R) operates on the binary question (default to retention); the displacement criterion (D) operates on the ordinal question (displace the lowest-ranked occupant). □

4.2 Inseparability and indexed knowledge

Corollary 4 (K/A Inseparability)**.** *The content of K is constitutively determined by A. Two entities $E_1 = (K_1, A_1, \sigma, \Omega_1)$ and $E_2 = (K_2, A_2, \sigma, \Omega_2)$ with identical sensing functions but different action spaces $A_1 \neq A_2$ will, in the curation regime, maintain different knowledge graphs $K_1 \neq K_2$.*

Proof. By Theorem 1, K contains the shared propositions across A. Different action spaces share different propositions: if $a \in A_1 \setminus A_2$, propositions exclusively grounding a have positive marginal value for E_1 but zero marginal value for E_2. Under the displacement criterion (Corollary 3(D)), E_2 displaces these propositions in favor of candidates relevant to A_2, while E_1 retains them. Provided $A_1 \setminus A_2$ contains actions whose grounding propositions are exclusive to those actions, $K_1 \neq K_2$.

The inseparability is at the *optimization* level: systems may implement K and A as distinct modules, but cannot optimize them independently. The training objective for K must reference A (which propositions are action-relevant), and vice versa (which actions are groundable given current K). □

4.3 Compactness and forgetting

Corollary 5 (Asymptotic Compactness)**.** *Under curation pressure, K converges toward the action-relevant subset $\bigcup_{a \in A} K^{[a]}$.*

Proof. A proposition p with no connection to any current or foreseeable action has zero estimated marginal value but positive maintenance cost $c_{\text{retain}}(p, K) > 0$. In the growth regime ($|K| < K_{\max}$), p is retained because certain zero benefit cannot be verified (Corollary 3(P)). In the curation regime, any candidate q connected to some $a \in A$ has strictly higher estimated marginal value than p, so p is displaced (Corollary 3(D)). Over time, propositions with no estimated action relevance are progressively displaced. The retained set converges toward $\bigcup_{a \in A} K^{[a]}$ asymptotically, with the rate depending on the accuracy of the entity's marginal-value estimates and the frequency of curation events. □

Corollary 6 (Curation and Forgetting)**.** *Under bounded K ($|K| = K_{\max}$), forgetting is correct behavior. Curation displaces propositions of lower marginal value when higher-value candidates arrive.*

Proof. By Corollary 3(D), displacement occurs when a candidate q has strictly higher net marginal value than the current occupant p. This is not information loss but information improvement: the entity's net survival benefit improves with each displacement. □

Remark (Free context). Define the *free context* $C_{\text{free}}(t) = C_n - |\text{ctx}_{\text{unresolved}}(t)|$: the portion of bounded context available for novel problems after existing knowledge maintenance. Every proposition retained in K reduces C_{free} by its retrieval interference and edge maintenance overhead. Curation under displacement (Corollary 3(D)) is therefore a C_{free}-maximizing operation: displacing low-value propositions frees context for higher-value candidates.

5 Strong Inseparability: The Encoding Continuum

Remark (K and A Are Not Natural Kinds). Given Theorem 1 (K precipitates from A) and the encoding hierarchy $E(c, r)$ of [14] (a continuous space parameterized by runtime cost c and reversibility r), K and A are not disjoint categories but regions on $E(c, r)$. The encoding hierarchy is empirically instantiated by the complementary learning systems of McClelland et al. [12]: hippocampus (high cost, high reversibility, fast episodic learning) and neocortex (low cost, low reversibility, slow statistical consolidation). Memory consolidation is movement along $E(c, r)$, operating in both directions.

Argument. We exhibit both directions of flow.

$K \to A$ **(declarative to procedural).** A proposition p grounding action a begins at encoding level L_n (active reasoning): explicitly represented, revisable. As the entity repeatedly applies a successfully, p migrates to lower encoding levels. At L_{n-1} (learned heuristic), p is retrievable but not explicitly stated. At L_1 (developmental encoding), p is compiled into the execution of a. At L_0 (heritable encoding), p is implicit in the wiring that produces a. This is the declarative-to-procedural transition observed in skill acquisition [10]: the novice follows explicit rules (K at L_n); the expert executes embodied competence (A at L_1).

$A \to K$ **(procedural to declarative).** Novel information often enters as pure action. A child touches a hot stove and pulls back. The first encoding is A: *don't touch that.* No proposition about temperature or tissue damage is represented. Only later does the entity derive the propositional structure: "the stove is hot," "hot things burn," "heat transfers from higher to lower temperature." Each proposition is derived from the action, not the other way around.

Bidirectionality. At no point in either direction does information cross a boundary from "knowledge" to "action." The encoding level determines whether information appears propositional (high c, high r) or procedural (low c, low r), but the information content is continuous through the hierarchy. The entity tuple $E = (K, A, \sigma, \Omega)$ is notational convenience: K is the high-encoding region of a single structure, A is the low-encoding region, and information flows in both directions.

Implication. The system does not "decide" whether information belongs in K or A. The encoding dynamics decide. Some information enters as explicit propositions and compiles into action ($K \to A$). Some enters as pure action and generates propositions only on reflection ($A \to K$). Some never becomes propositional at all: the expert's "feel" for a material, the athlete's timing. These are A without K: action knowledge that was never derived from and may never produce explicit propositions.

This strengthens the normalization result. The theorem proves the weak version: K separates from A as the normalized shared propositions. The continuum result proves the strong version: K and A are the same structure viewed at different encoding depths, with bidirectional flow. Separating them architecturally requires classifying each piece of information as "world model" or "action policy" at acquisition, but for $A \to K$ information, the classification is wrong at entry; for $K \to A$ information, the classification becomes wrong over time.

Remark on modularity of mind. The strongest version of the objection that K and A are natural kinds comes from Fodor's modularity thesis [7]. Fodor argues that mental modules are informationally encapsulated: they process input without access to information held by other modules. Carruthers [4] extends the argument to central cognition.

The encapsulation is real. The axis of encapsulation is not. Fodor's modules are encapsulated along the encoding hierarchy $E(c, r)$: a processor at L_1 cannot access representations at L_n. The visual module occupies a region of $E(c, r)$, not a side of a K/A partition. The same module holds both knowledge (edge statistics, color constancy priors) and action (saccade targeting, vergence control) at its encoding level. Encapsulation runs *between encoding levels*, not between knowledge and action. Physical modularity—hippocampus, neocortex, motor cortex, cerebellum—is $E(c, r)$ physically instantiated, not evidence for K/A as a natural partition.

6 Objections and Scope

6.1 The overlap assumption

The normalization theorem requires $\beta > 0$: at least some propositions are shared across action policies. Is this assumption restrictive?

In practice, $\beta > 0$ is the norm for any entity with a non-trivial action repertoire. Physical laws, spatial relations, object identities, social norms, and causal regularities ground multiple action policies simultaneously. An entity whose action policies share *no* propositions would have a purely reflexive repertoire—each action depending only on W_{obs}, with no propositional grounding at all. Such an entity has $K = \emptyset$ and the normalization question does not arise. For any entity with $K \neq \emptyset$ and $|A| > 1$, overlap is the expected condition.

6.2 The scope of survival pressure

The retention asymmetry (Corollary 3) depends on C_{absent} dominating all finite costs, which requires survival pressure. But most knowledge does not appear survival-critical.

The answer turns on what the entity can know *in advance* about which propositions are survival-critical. Under genuine uncertainty about future domains of encounter, the entity cannot partition its knowledge into "survival-relevant" and "not." Any proposition might ground a survival-critical action in an unforeseen domain. The proposition "Paris is the capital of France" is survival-irrelevant on an ordinary Tuesday and survival-critical when navigating an unfamiliar city to reach a hospital.

The cost of wrongly classifying a proposition as non-critical (C_{absent}) dominates the cost of wrongly classifying it as critical (c_{retain}), by the same asymmetry the corollary establishes. The conservative criterion therefore extends to all propositions whose future relevance is uncertain—which, for any finite entity, is nearly all propositions.

For entities genuinely not under survival pressure—archival databases, static knowledge stores—the retention asymmetry weakens. C_{absent} is finite, and the symmetric marginal value comparison of the curation regime becomes the primary decision rule.

6.3 The unfalsifiable pruning criterion

The pruning condition (P) requires $P(\exists\, \tau^*) = 0$: certain zero benefit across all reachable trajectories. No finite agent can verify this. Does this make pruning operationally vacuous?

The unfalsifiability is a feature, not a defect. It explains three empirical phenomena:

(1) Knowledge hoarding in the growth regime. Biological systems accumulate enormous quantities of seemingly useless information before capacity forces curation. This is the behavior Corollary 3 prescribes: in the growth regime, the unfalsifiable pruning criterion means *never voluntarily discard*.

(2) Forgetting is always forced, never voluntary. Biological forgetting is driven by capacity pressure rather than deliberate pruning [2]. The corollary explains why: the pruning condition cannot be verified, so forgetting happens only via displacement (D).

(3) The asymmetry operates within curation. The displacement criterion is not symmetric "higher value wins." The burden of proof falls on displacement: the challenger must be strictly better. Ties go to the incumbent.

7 Engagement with Prior Work

Codd [1]. The normalization theorem is the knowledge-system analogue of Codd's relational normalization. Codd identified that redundant storage of data across relations produces update anomalies and wasted capacity; normal forms extract shared dependencies into referenced tables. We prove that the same pressure operates on bounded cognitive systems: shared propositions across action policies produce redundant context load and update inconsistency; K is the normal form that eliminates both. The contribution is showing that normalization is not merely a good engineering practice but a necessary consequence of bounded context.

Dupoux, LeCun, and Malik [6]. Their proposed architecture separates System A (observation and world modeling) from System B (action and planning), with separate training objectives. Corollary 4 shows that separate training objectives violate a structural constraint: the content of the world model must reference the action space (which propositions are action-relevant), and the action space must reference the world model (which actions are groundable). Separate modules with a shared loss is compatible; separate training objectives is not.

Information Bottleneck [19, 20]. The information bottleneck retains information relevant to an output variable subject to a rate constraint, with threshold controlled by Lagrange multiplier β. Our retention criterion (Corollary 3) recovers a version of this trade-off but derives the threshold rather than positing it: under survival pressure, C_{absent} is bounded below by entity non-survival, producing a maximally conservative retention rule. The normalization theorem adds a structural result the information bottleneck lacks: not just *which* information to retain, but *how* to organize it (shared propositions separated from policies).

Active inference [8, 9]. Active inference argues that perception and action are jointly optimized under variational free energy minimization. This is the closest existing framework to K/A inseparability. What Theorem 1 adds is the organizational constraint under hard capacity limits. When context is unbounded, all information can be retained and the coupling is a soft preference. Under bounded C_n, the coupling becomes a hard constraint: propositions must be displaced, and the displacement criterion must reference the action space. Active inference tells the entity what to do; the normalization theorem tells it how to organize what it knows while doing it.

Gibson [11]. Gibson's affordances—properties of the environment specified relative to an organism's action capabilities—are knowledge indexed by action under a different name. Corollary 4 formalizes Gibson's insight. What the normalization theorem adds is a structural account: affordances are not merely perceived but *factored out* from action policies as shared propositions, and this factoring is forced by bounded capacity.

Anderson and Schooler [2]. They demonstrated that memory retention statistics mirror environmental statistics. This is an empirical prediction of Corollary 3: if retention is indexed by action relevance, and action relevance tracks environmental statistics, then retention curves should mirror environmental frequency distributions—precisely what they observe.

Simon [18]. The retention criterion is a satisficing criterion in Simon's sense. The entity does not compute the optimal knowledge set (a combinatorial problem) but applies a simple rule: retain if benefit is not certainly zero, displace if a better candidate arrives. When the benefit function is submodular, the greedy criterion achieves a $(1-1/e)$-approximation to optimal [16]. In the general case, the guarantee does not apply, but the criterion remains a well-defined satisficing heuristic.

8 Conclusion

Knowledge as normalization operates at two levels.

The *structural* level (Theorem 1): bounded active context with overlapping action policies forces the extraction of shared propositions into a separated structure K, converting multiplicative context load ($O(n \cdot k)$) to additive load ($O(n + k')$) and eliminating update anomalies. This is not a design choice but a scaling constraint. The entity that does not normalize cannot scale its action repertoire beyond C_n/k policies.

The *content* level (Corollaries 3–4): the propositions retained in K are constitutively determined by A. The retention criterion is asymmetric (retain when benefit is uncertain; prune only when certainly zero) and derived from the cost structure of bounded context under survival pressure, not from a design parameter. Systems may implement K and A as separate modules, but cannot give them independent training objectives without accepting information loss or reconstruction cost.

The *strong* inseparability (Remark 5): K and A are not natural kinds but regions on the encoding hierarchy $E(c, r)$, with information flowing in both directions. The boundary is not merely blurry; it is incoherent as a classification criterion.

The knowledge graph precipitates from the action space as the normalization of shared propositions. It is not a store of declarative facts separate from procedural skills; it is the shared structure underlying both, factored out to avoid redundancy under bounded capacity.

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References

[1] E. F. Codd. A relational model of data for large shared data banks. *Communications of the ACM*, 13(6):377–387, 1970.

[2] J. R. Anderson and L. J. Schooler. Reflections of the environment in memory. *Psychological Science*, 2(6):396–408, 1991.

[3] T. M. Cover and J. A. Thomas. *Elements of Information Theory*. Wiley-Interscience, 2nd edition, 2006.

[4] P. Carruthers. *The Architecture of the Mind: Massive Modularity and the Flexibility of Thought*. Oxford University Press, 2006.

[5] A. K. Dixit and R. S. Pindyck. *Investment Under Uncertainty*. Princeton University Press, 1994.

[6] E. Dupoux, Y. LeCun, and J. Malik. Why AI systems don't learn and what to do about it: Lessons on autonomous learning from cognitive science. *arXiv preprint arXiv:2603.15381*, 2026.

[7] J. A. Fodor. *The Modularity of Mind*. MIT Press, 1983.

[8] K. Friston. The free-energy principle: A unified brain theory? *Nature Reviews Neuroscience*, 11(2):127–138, 2010.

[9] T. Parr, G. Pezzulo, and K. J. Friston. *Active Inference: The Free Energy Principle in Mind, Brain, and Behavior*. MIT Press, 2022.

[10] P. M. Fitts and M. I. Posner. *Human Performance*. Brooks/Cole, 1967.

[11] J. J. Gibson. *The Ecological Approach to Visual Perception*. Houghton Mifflin, 1979.

[12] J. L. McClelland, B. L. McNaughton, and R. C. O'Reilly. Why there are complementary learning systems in the hippocampus and neocortex: Insights from the successes and failures of connectionist models of learning and memory. *Psychological Review*, 102(3):419–457, 1995.

[13] P. D. McCarthy. Graph structure is necessary for information preservation under bounded context. Technical report, 2026.

[14] P. D. McCarthy. Convergence of control architectures to escalation under bounded context. Working paper, 2026.

[15] P. D. McCarthy. Evaluation-driven descent, encoding permanence, and the structural invariants of intelligence. Manuscript, 2026.

[16] G. L. Nemhauser, L. A. Wolsey, and M. L. Fisher. An analysis of approximations for maximizing submodular set functions — I. *Mathematical Programming*, 14(1):265–294, 1978.

[17] C. E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27(3):379–423, 1948.

[18] H. A. Simon. A behavioral model of rational choice. *The Quarterly Journal of Economics*, 69(1):99–118, 1955.

[19] N. Tishby, F. C. Pereira, and W. Bialek. The information bottleneck method. In *Proceedings of the 37th Annual Allerton Conference on Communication, Control, and Computing*, pages 368–377, 2000.

[20] N. Tishby and N. Zaslavsky. Deep learning and the information bottleneck principle. In *IEEE Information Theory Workshop*, pages 1–5, 2015.

[21] P. D. McCarthy. Coupling channels, severance hierarchy, and the prognostic structure of cancer: an empirical test of bounded-context theory. *SSRN preprint*, 2026.